

AKNLTP014_REMED1_RCB Python vs Rust Parity Report

Input: /Users/arunsasidharan/EEGdata/EEGAnalysisCode/sampleData/AKNLTP014_REMED1_RCB.set

Channels: Fz, F3, F4, C3, C4, P3, P4, Pz, O1, O2

Window: first 15 seconds, epoch length 15 seconds

Feature Extraction

Python feature shape: [10, 111]

Rust feature shape: [10, 111]

Common numeric features compared: 104

Family summary is in feature_comparison_by_family.csv; worst columns are in feature_comparison_by_column.csv.

Family	Features	Max abs	Median mean abs	Median corr
acw	1	0	0	1
connectivity	54	0.67517	2.68137e-09	1
foeof	19	5.57945	0.000710271	0.99998
irasa	12	1.33674e-06	2.24617e-08	1
metadata	3	0	0	NA
nonlinear	8	0.000319949	6.53261e-10	1
psd	7	2.05291e-08	3.89176e-09	1

Worst Feature Columns

Feature	n	Max abs	Mean abs	Max rel	Corr
auc_FOOOF	10	5.57945	1.40531	8.17315	0.994337
conn_mic_Alpha	10	0.67517	0.67517	2	NA
conn_mic_Gamma1	10	0.364188	0.364188	15.3529	NA
cf_0_FOOOF	10	0.244197	0.0413328	0.0113035	0.999949
conn_mic_Theta	10	0.220766	0.220766	2.49628	NA
conn_mic_Beta1	10	0.218856	0.218856	1.50823	NA
conn_mic_ThetaAlpha	10	0.216948	0.216948	0.586956	1
conn_mic_Beta2	10	0.125749	0.125749	1.07784	1
conn_wp1i_Gamma1	10	0.0867167	0.0412525	0.410777	0.781987
cf_1_FOOOF	5	0.0850479	0.0412943	0.00385287	0.999999
conn_pli_Gamma1	10	0.0588467	0.0316493	0.634334	0.662396
conn_ciplv_Gamma1	10	0.0432772	0.025926	0.645928	0.685327
conn_mim_Gamma1	10	0.0399641	0.0399641	0.188131	NA
bw_0_FOOOF	10	0.0394723	0.00694913	0.00948375	1
conn_plv_Gamma1	10	0.0353942	0.0181362	0.140687	0.891935

Preprocessing

Python sample rate: 250.0

Rust sample rate: 250.0

Channels compared: Fz, F3, F4, C3, C4, P3, P4, Pz, O1, O2

Samples compared: 1000

RMS difference (uV): 4.33829

Median absolute difference (uV): 2.8171

Max absolute difference (uV): 22.1237

Snapshots:

- preprocess_snapshot_stacked.png
- preprocess_snapshot_Fz.png and preprocess_overlay_Fz.png
- preprocess_snapshot_F3.png and preprocess_overlay_F3.png
- preprocess_snapshot_F4.png and preprocess_overlay_F4.png
- preprocess_snapshot_C3.png and preprocess_overlay_C3.png
- preprocess_snapshot_C4.png and preprocess_overlay_C4.png
- preprocess_snapshot_P3.png and preprocess_overlay_P3.png
- preprocess_snapshot_P4.png and preprocess_overlay_P4.png
- preprocess_snapshot_Pz.png and preprocess_overlay_Pz.png

Visual summaries:

- feature_family_parity.png
- feature_scatter_parity.png
- connectivity_metric_parity.png

Interpretation

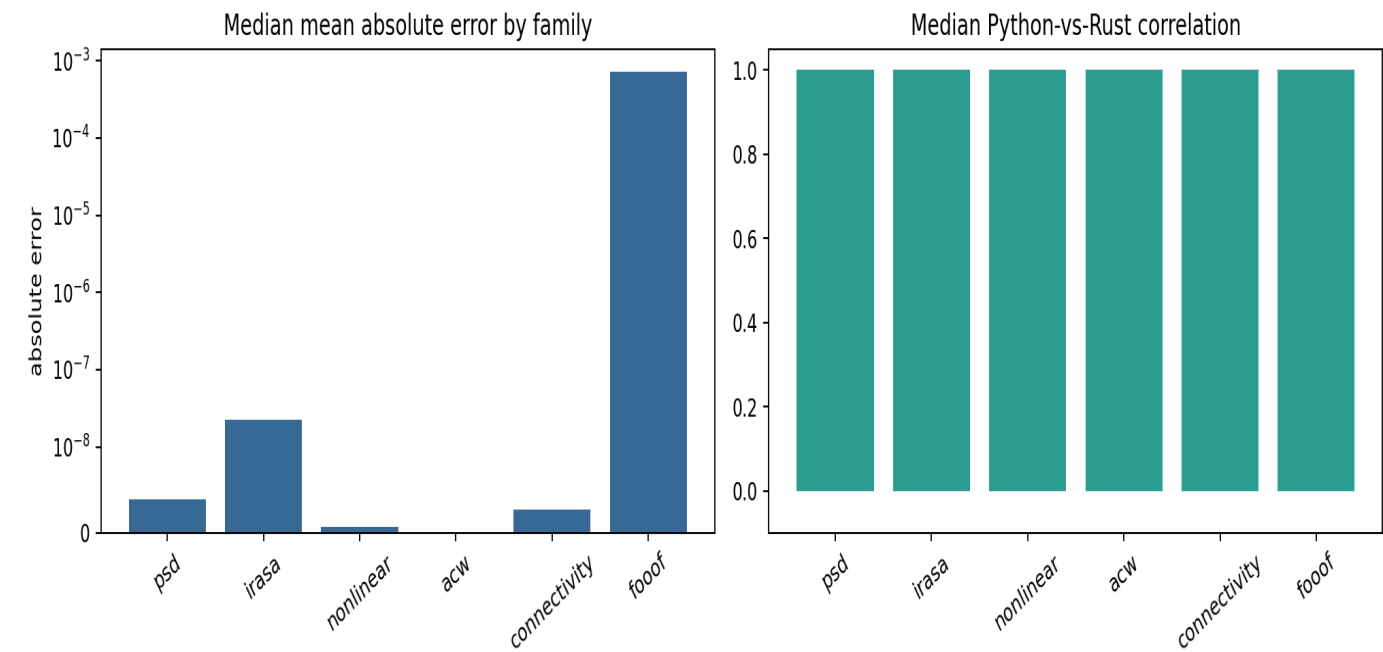
PSD, nonlinear features, ACW, and most connectivity values are at or near numerical precision against the Python reference on this real-data slice.

IRASA oscillatory band powers match closely; IRASA fitted aperiodic parameters remain less exact because Rust and Python fitting paths are not identical.

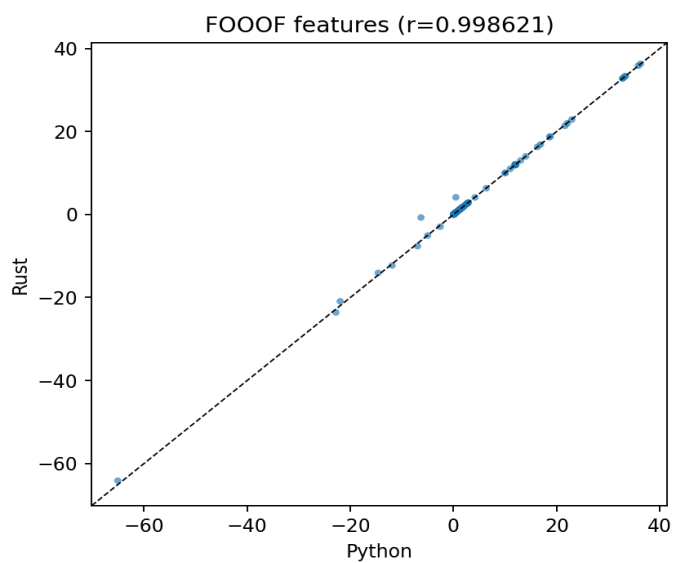
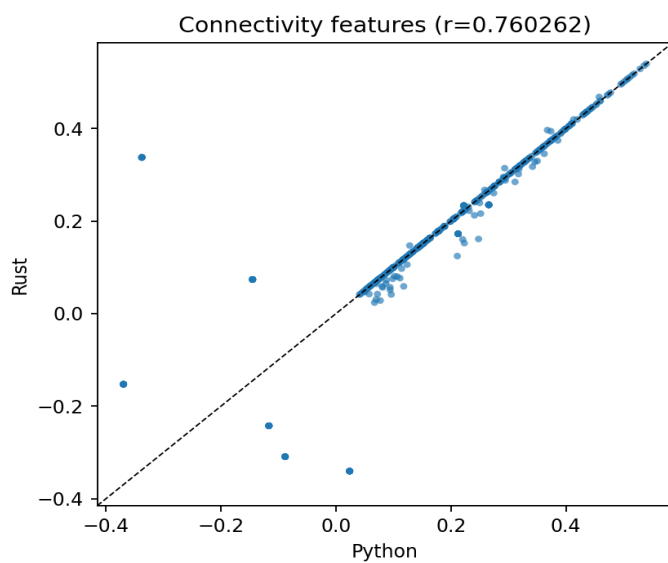
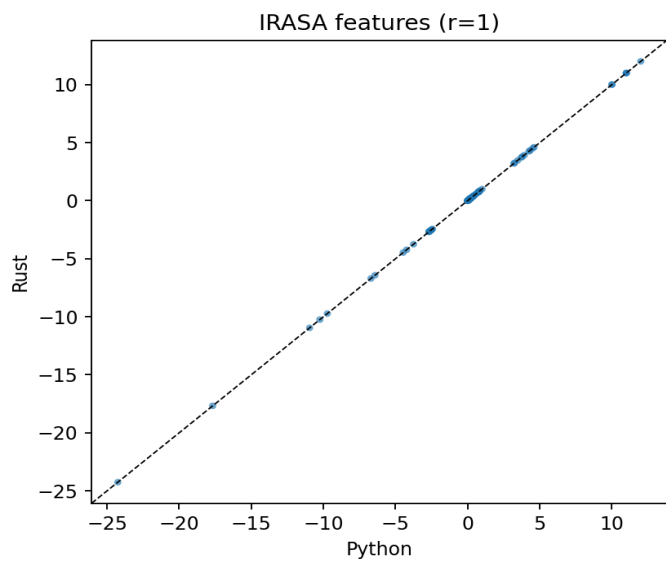
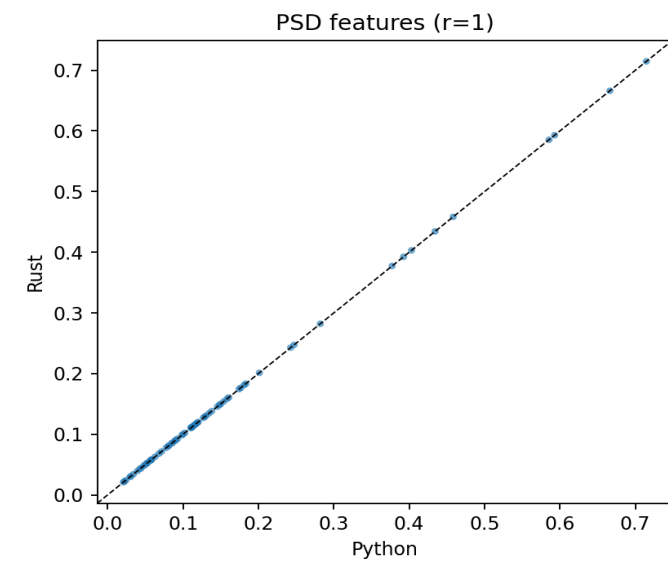
FOOOF and preprocessing remain the hardest native parity targets. The report includes scatter plots and shared-scale preprocessing overlays so achieved parity and remaining gaps are visible rather than only tabular.

Feature Parity Plots

feature_family_parity.png

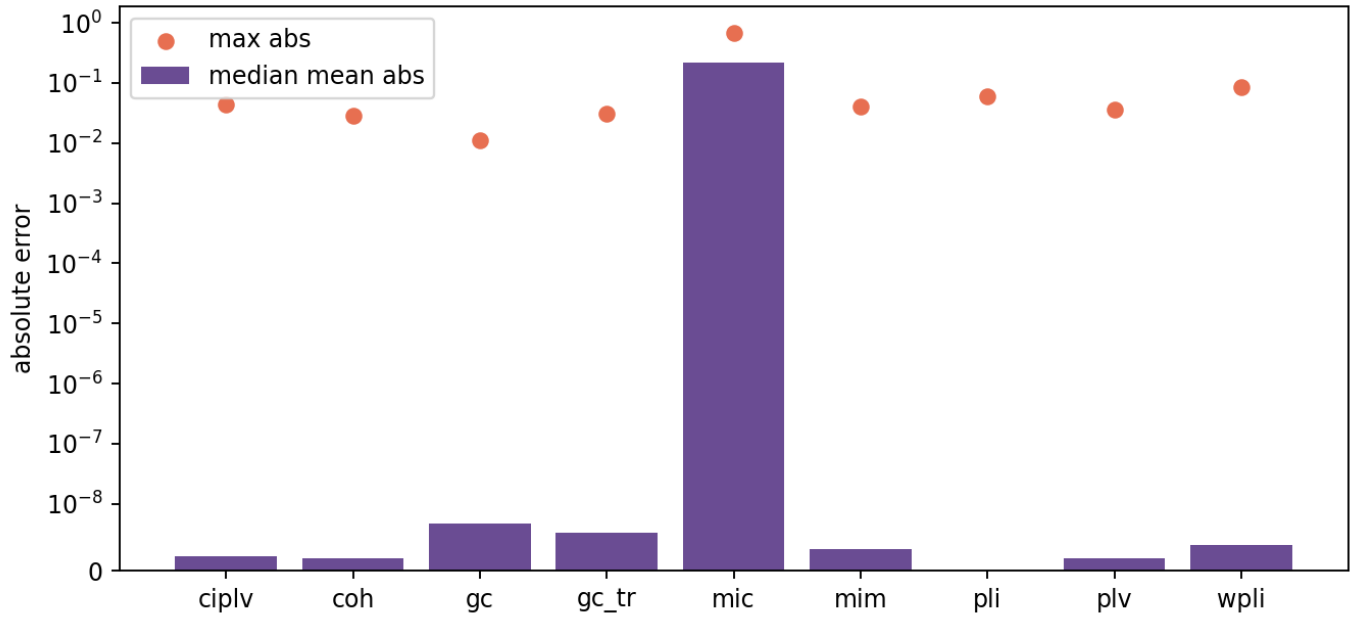


feature_scatter_parity.png



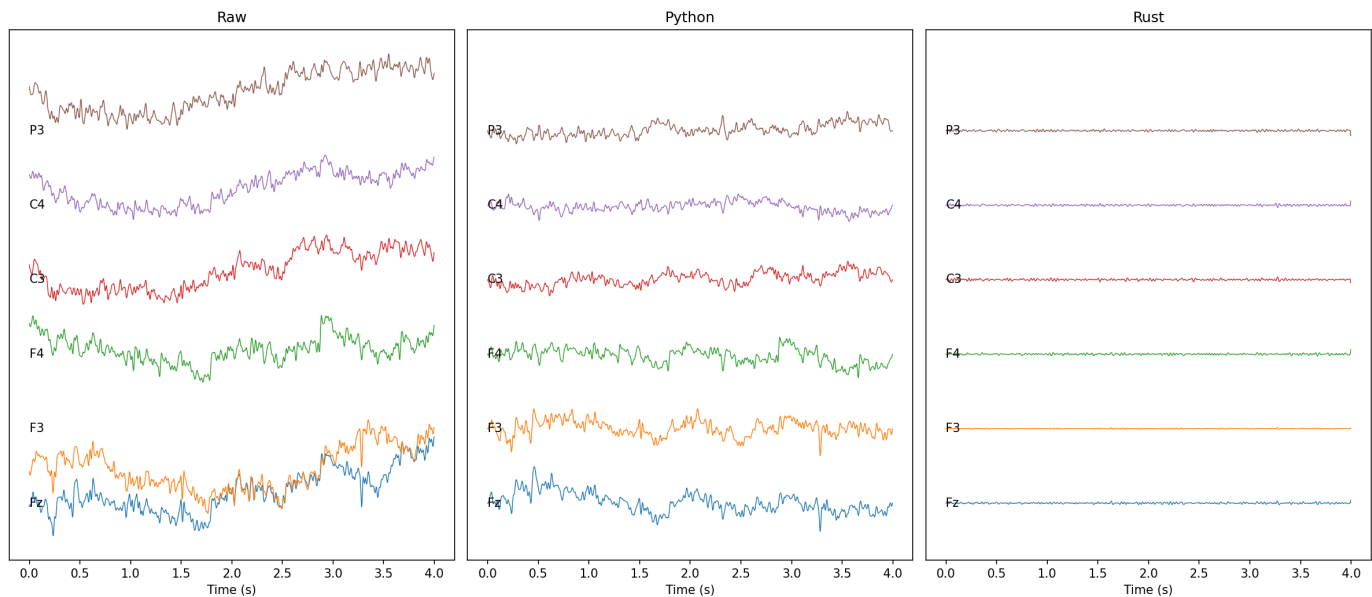
connectivity_metric_parity.png

Connectivity parity by metric

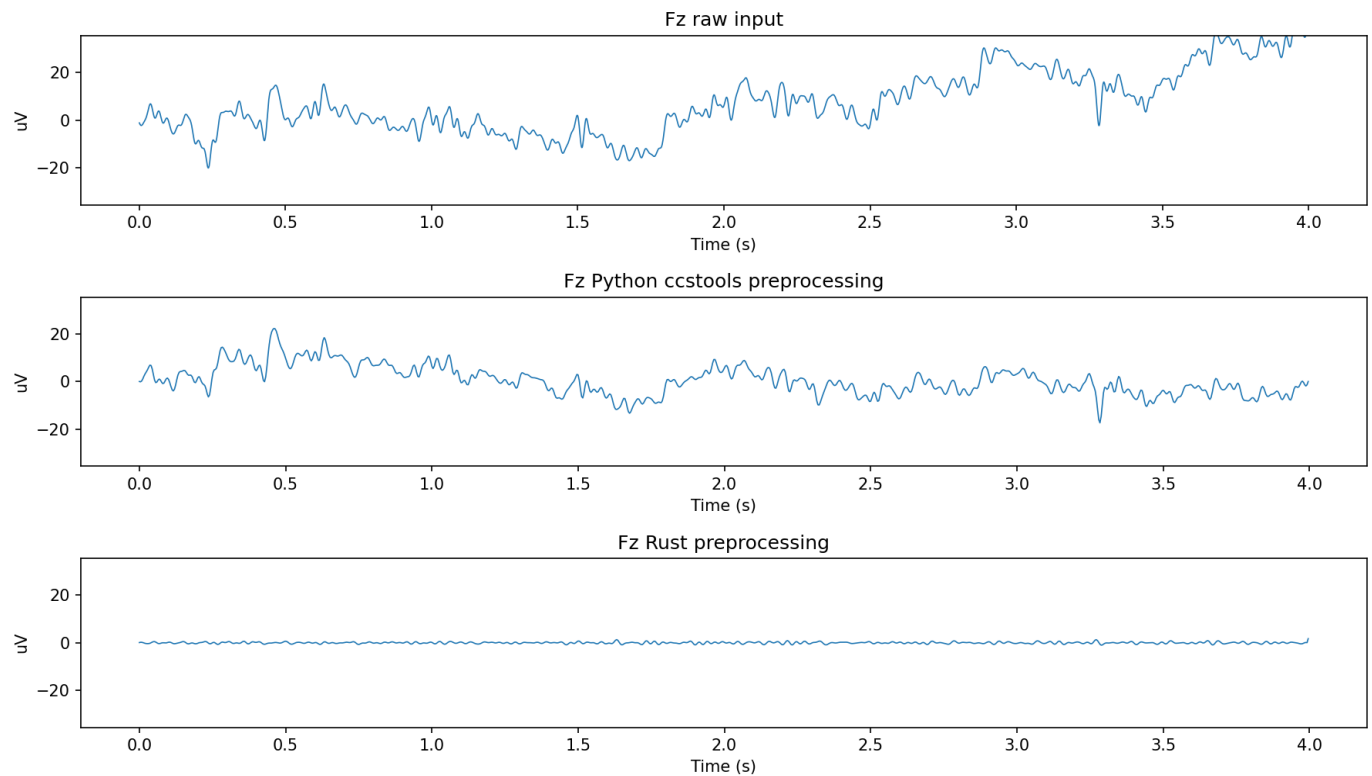


Preprocessing Snapshots

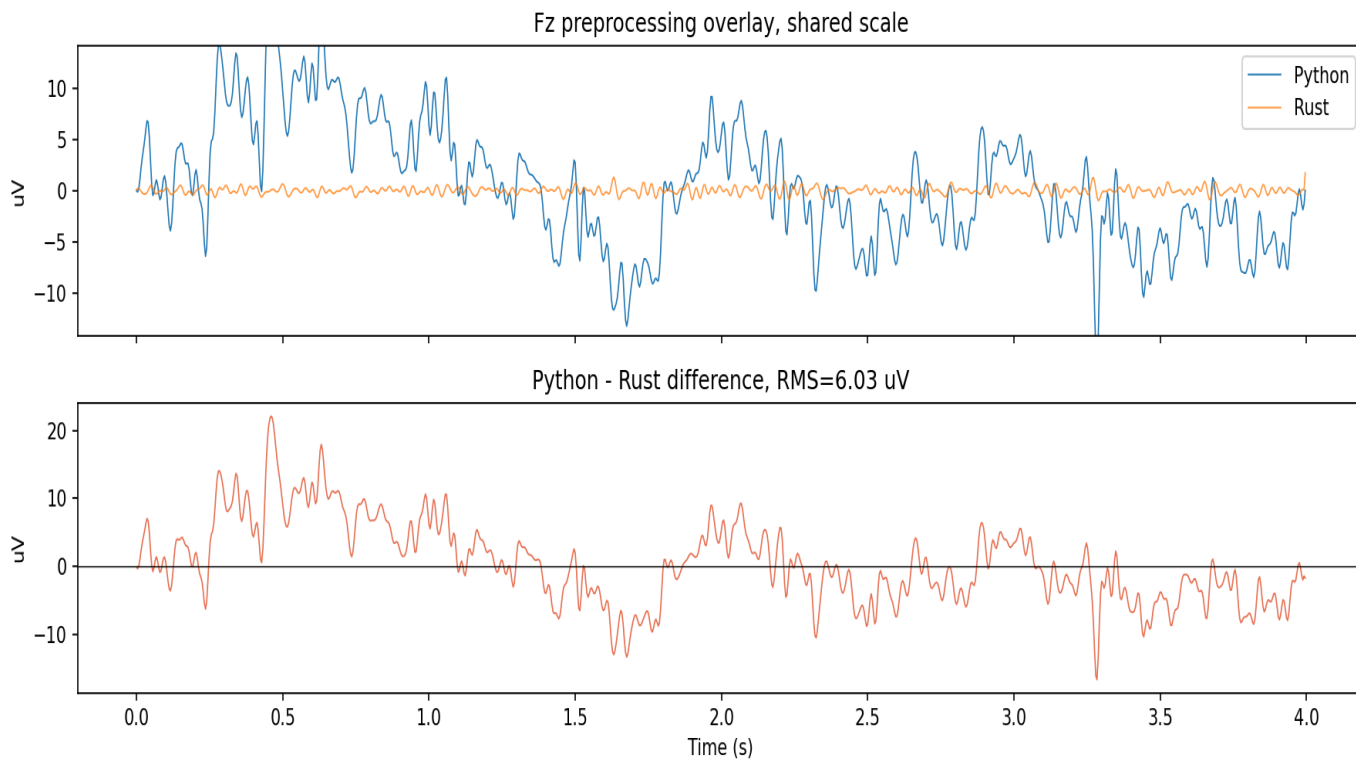
preprocess_snapshot_stacked.png



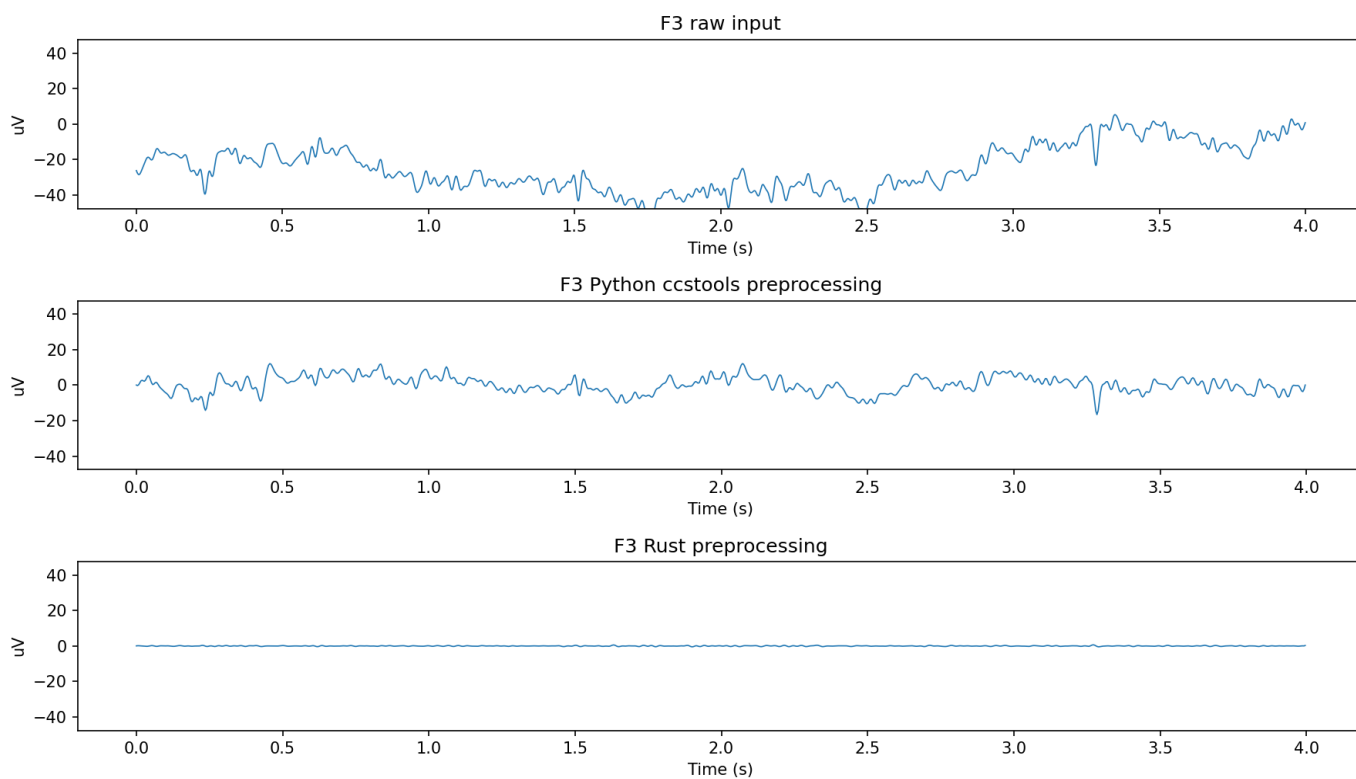
preprocess_snapshot_Fz.png



preprocess_overlay_Fz.png

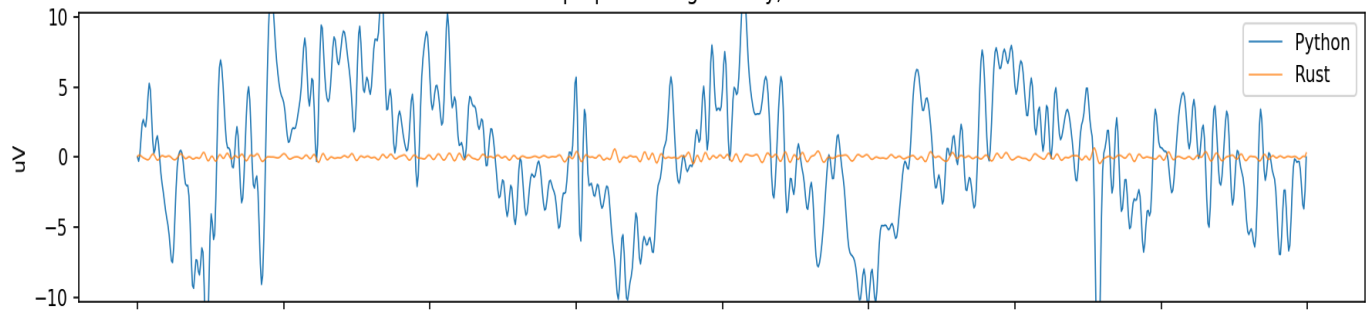


preprocess_snapshot_F3.png



preprocess_overlay_F3.png

F3 preprocessing overlay, shared scale



Python - Rust difference, RMS=4.63 uV

